sumup®

Home test

Preventing fraud in PIX transactions

Pix is one of the largest payment methods in Brazil. Unfortunately, many fraudsters take advantage of this to commit illegal acts and steal money from people. Since one of the SumUp's main goals is to grow our bank usage, we need to prevent our product from being used by fraudsters. Currently, any suspicious PIX transaction needs to be manually analyzed by our fraud prevention team. **But how can we identify which transactions they should evaluate? How do we select the best cases to be analyzed, considering we have limited human capacity?**

Your goal is to create a solution that involves machine learning to identify fraudulent transactions when a PIX is received by our merchant accounts (companies that have an account with SumUp) so that they can be analyzed by our fraud prevention team.

Source:

- *sumup_cas.parquet*: Historical PIX transaction data
- *DS_II_home_test_features.csv*: Features description (you don't need to use all features, if you don't want)

Deliverables:

- Machine learning model and other possible extra solutions
- API for model inference (remembering that the inference of each case should be executed at the same time the transaction occurs)
- All code/Notebook/etc. used in development
- At least one (1) slide presenting the solution created

Evaluation:

- To evaluate your thought process, we will assess your ability to work through the problem in a structured and logical way. We will look for evidence of your ability to identify the key elements of the problem, such as the data, the business context, and the goals of the project.
- We will evaluate your analytical and problem-solving skills, which includes your ability to apply and understand the appropriate statistical methods and machine learning algorithms to the problem.
- Finally, we will also evaluate your communication skills, looking for evidence of your ability to communicate your findings and recommendations clearly.

Deadline: 5 days from receipt of the case